

Python Practicum Objectives

In the capstone to the Python and Data track, teams of four students deploy an ETL pipeline to the cloud. The practicum is split into two phases: a local phase where students build their pipeline with Prefect + Supabase, then a cloud phase where they refactor to deploy using AWS Lambda.

PBL Approach

Driving Question: How do you take a data pipeline off your laptop and make it run on its own in the cloud, reliably and for free?

Each sprint opens with a mini-lesson (or two mini-lessons, in sprint 4), which introduces the new cloud content.

Dataset Options

1. Air Quality – OpenWeather
 2. Earthquakes – USGS
 3. Bike Share – GBFS
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Sprint Structure

Sprint 1 (Weeks 1-2)

Goal: The team is set up and pulling data.

Mini-Lesson: Working as a data team, contracts & the PR workflow

Must:

- Git and pull-request workflow; team working agreement
- Agreed data contracts (raw and gold schemas, function signatures)
- `extract()` client with tests

Should:

- Caching for any lookups (geocoding, station metadata)
- Architecture diagram refined

Could:

- A second data source scoped for later
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Sprint 2 (Weeks 3-4)

Goal: TBD

Mini-Lesson: The raw→gold pattern & idempotent writes

Must:

- Supabase schema (raw, gold, logs)
- Idempotent `load_raw()`
- `transform()` to gold — a category and a score

Should:

- Rolling aggregates and the data dictionary

- Run tracking in pipeline_runs

Could:

- Richer derived fields
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Sprint 3 (Weeks 5-6)

Goal: TBD

Mini-Lesson: Backfill vs. incremental — why a pipeline runs two ways

Must:

- LLM enrichment (incremental, idempotent)
- Prefect @flow with retries and logging
- A minimal Streamlit dashboard

Should:

- The ML classifier trained and integrated (the double-transform)

Could:

- Dashboard polish; extra charts
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Sprint 4 (Weeks 7-8)

Goal: TBD

Mini-Lessons:

1. **Week 7:** Serverless + setting up a cloud account safely (IAM, Budgets, SSM; the "ephemeral, triggered compute" model)
2. **Week 8:** Packaging & deploying a Lambda; scheduling with EventBridge

Must:

- AWS account and safety rails (Budgets alarm, limited IAM, SSM secrets)
- Ingest Lambda (extract + load_raw + transform) deployed
- EventBridge schedule and CloudWatch logs

Should:

- LLM enrichment as a second Lambda
- A basic CloudWatch alarm

Could:

- CI/CD deploy via GitHub Actions
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Sprint 5 (Weeks 9-10)

Goal: TBD

Mini-Lessons: Observability: logs, alarms, dead-letter queues (+ optional CI/CD)

Must:

- End-to-end run on AWS
- Cost-teardown check
- Before/after writeup, final demo and retrospective

Should:

- Anything deferred from Sprint 4 (enrichment Lambda, alarm)
- SNS or Slack failure alert

Could:

- Dead-letter queue
- ML classifier in a container Lambda
- Infrastructure as Code (AWS SAM)